

# A CAYLEY GRAPH ON $\mathbb{Z}_{15}$ WHOSE CORE IS $C_5$

ARGUS OPEN RESEARCH PROJECT

**ABSTRACT.** We give an explicit connected Cayley graph on the cyclic group  $\mathbb{Z}_{15}$  whose core is the five-cycle. Let

$$X = \text{Cay}(\mathbb{Z}_{15}, \{1, 4, 6, 9, 11, 14\}).$$

Reduction modulo 5 maps  $X$  homomorphically onto  $C_5$ , and the map  $i \mapsto 6i$  gives a section whose image is an induced five-cycle. The resulting composite is a retraction. Every endomorphism of  $C_5$  is an automorphism, so  $C_5$  is the core of  $X$ . Since a Cayley graph on  $(\mathbb{Z}_{15})^m$  has  $15^m$  vertices, this core is not a Cayley graph on any power of  $\mathbb{Z}_{15}$ . The example refutes the literal arbitrary-abelian formulation of a problem on cores of Cayley graphs, but does not address prime, elementary-abelian, or cubelike variants.

## 1. THE GRAPH

Let

$$G = \mathbb{Z}_{15}, \quad \pi : G \longrightarrow \mathbb{Z}_5, \quad \pi(x) = x \bmod 5,$$

and put

$$S = \pi^{-1}(\{1, -1\}) = \{1, 4, 6, 9, 11, 14\}.$$

Define the simple undirected Cayley graph

$$X = \text{Cay}(G, S)$$

by declaring  $x$  adjacent to  $y$  precisely when  $y - x \pmod{15}$  lies in  $S$ .

The set  $S$  does not contain zero and satisfies  $S = -S$ , so  $X$  is a simple undirected graph. It is 6-regular and has

$$\frac{15 \cdot 6}{2} = 45$$

edges. Since  $1 \in S$ , the edges  $x \sim x + 1$  contain a spanning fifteen-cycle, and hence  $X$  is connected.

## 2. A RETRACTION ONTO AN INDUCED FIVE-CYCLE

Write

$$C_5 = \text{Cay}(\mathbb{Z}_5, \{1, -1\})$$

and define

$$\sigma : \mathbb{Z}_5 \longrightarrow \mathbb{Z}_{15}, \quad \sigma(i) = 6i \bmod 15.$$

Since  $6 \equiv 1 \pmod{5}$ , one has

$$\pi \circ \sigma = \text{id}_{\mathbb{Z}_5}.$$

**Lemma 2.1.** *The maps  $\pi : X \rightarrow C_5$  and  $\sigma : C_5 \rightarrow X$  are graph homomorphisms.*

*Proof.* If  $x \sim_X y$ , then

$$\pi(y) - \pi(x) = \pi(y - x) \in \{1, -1\},$$

so  $\pi(x) \sim_{C_5} \pi(y)$ .

Conversely, if  $i \sim_{C_5} j$ , then  $j - i = \pm 1$  in  $\mathbb{Z}_5$ . Therefore

$$\sigma(j) - \sigma(i) = \pm 6 \in S,$$

and hence  $\sigma(i) \sim_X \sigma(j)$ . □

The image of  $\sigma$ , in cyclic order, is

$$H = \{0, 6, 12, 3, 9\}, \quad 0 - 6 - 12 - 3 - 9 - 0.$$

---

*Date:* August 30, 2026.

*2020 Mathematics Subject Classification.* Primary 05C60; Secondary 05C25, 20K01.

*Key words and phrases.* Cayley graph, graph core, graph homomorphism, retraction.

**Lemma 2.2.** *The subgraph of  $X$  induced by  $H$  is a five-cycle, and*

$$R = \sigma \circ \pi : X \longrightarrow H$$

*is a retraction.*

*Proof.* For distinct  $i, j \in \mathbb{Z}_5$ ,

$$\sigma(j) - \sigma(i) = 6(j - i) \in S \iff j - i \equiv \pm 1 \pmod{5}.$$

Thus the only edges among the five image vertices are the five consecutive cycle edges, so  $H$  is induced. By Lemma 2.1, the composite  $R$  is a graph homomorphism. For  $h = \sigma(i) \in H$ ,

$$R(h) = \sigma(\pi(\sigma(i))) = \sigma(i) = h,$$

so  $R$  fixes  $H$  pointwise.  $\square$

### 3. THE FIVE-CYCLE IS A CORE

A finite graph is a *core* if every endomorphism is an automorphism.

**Lemma 3.1.** *Every endomorphism of  $C_5$  is an automorphism.*

*Proof.* Label the vertices by  $\mathbb{Z}_5$ , and let  $f : C_5 \rightarrow C_5$  be an endomorphism. Since consecutive vertices must map to adjacent vertices,

$$f(i+1) - f(i) = \epsilon_i, \quad \epsilon_i \in \{1, -1\}.$$

Summing over the five edges gives

$$\sum_{i=0}^4 \epsilon_i \equiv 0 \pmod{5}.$$

The sum of five signs is one of

$$-5, -3, -1, 1, 3, 5.$$

Only  $\pm 5$  is divisible by 5, so all signs are equal. Consequently  $f(i) = a + i$  for all  $i$ , or  $f(i) = a - i$  for all  $i$ . These are the rotations and reflections of  $C_5$ , and each is an automorphism.  $\square$

**Proposition 3.2.** *The core of  $X$  is isomorphic to  $C_5$ .*

*Proof.* By Lemma 2.2,  $X$  retracts onto  $H \cong C_5$ . Hence  $X$  and  $C_5$  are homomorphically equivalent. By Lemma 3.1,  $C_5$  is already a core. The core is unique up to isomorphism within a finite homomorphic-equivalence class.  $\square$

### 4. THE POWER-CARDINALITY OBSTRUCTION

**Theorem 4.1.** *There is a Cayley graph on  $\mathbb{Z}_{15}$  whose core is not a Cayley graph on any power of  $\mathbb{Z}_{15}$ .*

*Proof.* Take the graph  $X$  above. Its core has five vertices by Proposition 3.2. A Cayley graph on  $(\mathbb{Z}_{15})^m$  has exactly  $15^m$  vertices. If  $m = 0$  is allowed, this number is 1; if  $m \geq 1$ , it is divisible by 3. In neither case is it 5.  $\square$

**Remark 4.2** (Scope). The example uses the composite-order cyclic group  $\mathbb{Z}_{15}$ . It refutes an unrestricted formulation quantifying over arbitrary abelian groups. It does not refute variants restricted to  $\mathbb{Z}_p$ , elementary abelian groups, or cubelike graphs.

### APPENDIX A. CHECKLIST FOR VERIFICATION

- (1) Check that  $S = -S$ ,  $0 \notin S$ , and  $1 \in S$ .
- (2) Verify that reduction modulo 5 preserves every edge.
- (3) Verify that  $i \mapsto 6i$  preserves every edge of  $C_5$ .
- (4) Check that the image  $\{0, 6, 12, 3, 9\}$  is induced and that the composite fixes it pointwise.
- (5) Verify the sign-sum proof that every endomorphism of  $C_5$  is an automorphism.
- (6) Compare 5 with  $15^m$  for every  $m \geq 0$ .
- (7) Keep the arbitrary-abelian statement separate from the prime and elementary-abelian variants.

## REFERENCES

- [1] P. Hell and J. Nešetřil, *Graphs and Homomorphisms*, Oxford Lecture Series in Mathematics and its Applications, vol. 28, Oxford University Press, 2004.
- [2] R. Šámal, *Cores of Cayley graphs*, Open Problem Garden, Problem 125 (2008), with a counterexample observation by G. Royle and subsequent scope clarification by the author.

ARGUS OPEN RESEARCH PROJECT